

MUHAMMAD FARHAN

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Mid-to-Senior Backend Developer with 4 years of experience designing and building scalable web applications, high-performance APIs, and event-driven distributed systems. Highly proficient in Node.js (TypeScript and JavaScript) and Go, with deep practical expertise in Docker, Kubernetes, RabbitMQ, and Redis. Proven track record in microservices architecture, system modernization (achieving up to 75% resource reduction), and financial/blockchain gateway engineering. Maintains a strong foundational computing background with a 4.0 GPA in Computer Science (AI Specialization).

EXPERIENCE

BACKEND DEVELOPER

PT Inovasi Solusi Internasional | Jakarta, Indonesia

2022 – PRESENT

- a. Regol (Forex Client Onboarding System)
 - Spearheaded the complete backend migration of a high-traffic Forex client registration platform from legacy PHP to TypeScript.
 - Identified and resolved critical legacy bugs, ensuring 100% functional parity with the previous system while enhancing codebase reliability.
 - Optimized system performance, successfully reducing overall server resource utilization by up to 75% during off-peak hours.
- b. Eform (Internal CRM & Customer Management)
 - Executed the strategic rewrite of an internal CRM from PHP to a TypeScript/Express.js microservices architecture, deploying standard Docker environments to achieve zero-downtime migration.
 - Mitigated numerous security vulnerabilities and bugs identified in penetration testing reports, significantly hardening the application's security posture.
 - Implemented a dynamic, granular Role-Based Access Control (RBAC) system allowing clients to configure permissions via a dashboard, which included automated data masking/redaction on sensitive documents for unauthorized roles.
 - Engineered the migration with zero downtime, introduced new client-requested features, and achieved a 30% reduction in computing resource usage.
- c. Dynamic Object Detection System
 - Developed a dynamic Computer Vision system leveraging Deep Learning with YOLOv8 for real-time object detection and automated bounding box rendering.
 - Designed a flexible video ingestion pipeline supporting multiple, dynamic input sources switched on-the-fly by users, including remote CCTV streams, direct server webcams, and client-side webcams.
- d. Etrade (Forex Trading & Microservices Architecture)
 - Collaborated on rebuilding a core Forex trading system from scratch using Express.js and JavaScript under strict constraints with zero access to the legacy vendor's locked source code or database.
 - Designed and containerized multiple independent microservices using Docker and Kubernetes, leveraging RabbitMQ for asynchronous event-driven communication to ensure high system resilience.
 - Optimized data aggregation pipelines by implementing strategic Redis caching and query tuning, successfully slashing analytics response times from 30s to under 3s.
 - Built entirely new system modules and successfully modernized a legacy desktop-based exchange reporting application into a scalable web platform.
- e. Blockchain Payment Gateway

- Engineered a dynamic multi-chain crypto payment gateway supporting USDT across Ethereum, Binance Smart Chain (BSC), Tron, and other networks.
- Developed automated on-chain verification mechanisms that monitor transaction confirmations and block depths to prevent double-spending and ensure payment finality.
- Created an abstract configuration layer allowing dynamic fee management and seamless addition of new blockchains or tokens, delivering a frictionless user experience akin to QRIS scanning.

EDUCATION

Binus University (Online Learning) – Computer Science

AUGUST 2023 - PRESENT

- Current Status: 6th Semester | GPA: 4.0 / 4.0
- Specialization (Streaming): Artificial Intelligence
- Core Computer Science coursework: Algorithm & Programming Foundations, Operating Systems (Threads & CPU Cycles), Discrete Mathematics, Calculus, and Dynamic Programming.
- AI & Data Science coursework: Natural Language Processing (NLP), Machine Learning, Computer Vision Applications, and Data Science.

SKILLS

- Programming Languages: JavaScript, TypeScript, Go (Golang), SQL, HTML/CSS.
- Backend Development & Frameworks: Node.js, Express.js, Go-Chi, Gin Gonic, Microservices Architecture, GRPC, Event-Driven Development, RESTful APIs, Role-Based Access Control (RBAC).
- Database & Tools: PostgreSQL, MySQL, Redis, ElasticSearch.
- DevOps & Infrastructure: Git, Linux/Unix Environments, Docker, Kubernetes, RabbitMQ.
- Core Competencies: System Modernization & Migration, Database Optimization, Security Hardening & Vulnerability Mitigation, Reverse Engineering, Agile/Scrum Development, Technical Problem Solving, Cross-functional Collaboration.
- Productivity Tools: Claude Code, Antigravity CLI